

An ICN-Based Secure Task Cooperation in Challenging Wireless Edge Networks

Ningchun Liu¹, Shuai Gao, *Member, IEEE*, Xindi Hou, Teng Liang, Guobiao He²,
Hongke Zhang³, *Fellow, IEEE*, and Sajal K. Das⁴, *Fellow, IEEE*

Abstract—Task cooperation emerges as an efficacious strategy for the execution of intricate tasks within the context of challenging wireless edge networks characterized by limited resources and intermittent infrastructure connections. Presently, TCP/IP-based solutions encounter issues related to suboptimal utilization of network resources and a substantial dependency on infrastructure connections. In light of this, Information-Centric Networking (ICN) has surfaced as a promising architectural paradigm aimed at mitigating these challenges. In ICN-based task cooperation, the data reuse characteristic of ICN enhances the efficiency of network resource utilization. However, this also introduces plausible security vulnerabilities to the reused data, encompassing eavesdropping attacks and unauthorized access attacks. In this paper, we propose an ICN-based secure task cooperation scheme to mitigate the above threats without compromising the efficiency of task execution. We present the task cooperation model that quantifies the cost of securing data reuse in task execution. We also introduce a specific naming convention to support the acquisition of collaborative task requests and keys related to the task cooperation. Besides, we introduce a novel design for an enhanced name-based access control scheme that ensures both data confidentiality and access control in collaborative tasks accessed by the same sub-policy, streamlining the encryption process for content keys. Security analysis and experimental results demonstrate that our scheme effectively safeguards the security of data reuse. Furthermore, compared to existing schemes, our scheme incurs lower cost associated with computation and security.

Index Terms—ICN, secure task cooperation, wireless edge networks.

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Ningchun Liu is with the Research Institute of Intelligent Networks, Zhejiang Laboratory, Hangzhou 311121, China, and also with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, China.

Shuai Gao, Xindi Hou, and Hongke Zhang are with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, China (e-mail: shgao@bjtu.edu.cn).

Teng Liang is with the Department of New Networks, Peng Cheng Laboratory, Shenzhen 518066, China.

Guobiao He is with the National Computer Network Emergency Response Technical Team/Coordination Center of China, Beijing 100029, China.

Sajal K. Das is with the Department of Computer Science, Missouri University of Science and Technology, Rolla, MO 65409 USA.

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I. INTRODUCTION

RECENTLY, unmanned vehicles have increasingly been utilized in various challenging scenarios, including disaster relief and military operations that involve intelligence, surveillance, and reconnaissance [1]. Considering the complexity of the task, task cooperation among multiple unmanned vehicles with limited resources within the wireless edge network is necessary in the aforementioned scenario [2]. Nevertheless, the majority of wireless edge networks currently rely on the TCP/IP network architecture, which presents several limitations in challenging scenarios characterized by intermittent connectivity, limited computational resources, and vulnerability to security threats. Precisely, the TCP/IP network architecture, initially crafted for static hosts and routers linked via wired connections, faces challenges in effectively utilizing the broadcast capabilities inherent to wireless channels [3]. Moreover, the network infrastructure in challenging scenarios might experience partial disablement or intermittent availability. In such contexts, the existence of a continuously connected path between source and destination cannot be assured, rendering traditional client-server protocols inadequate for supporting communication [4].

Positioned as a propitious network architecture, ICN [5] embraces the content-centric communication model and facilitates asynchronous content transmission between nodes without necessitating a continuous end-to-end connection. This characteristic deems it a fitting solution for tackling the aforementioned challenges within intermittently connected demanding scenarios [6]. Furthermore, owing to its inherent in-network caching capability, ICN can significantly amplify resource utilization efficiency in wireless edge networks by enabling efficient data reuse [7].

Numerous ICN-based schemes [7], [8], [9], [10], [11] have emerged to enhance network resource utilization in wireless edge networks through data reuse. Nevertheless, these schemes tend to overlook the critical imperative of safeguarding reused data, exposed to potential security vulnerabilities like eavesdropping attacks and unauthorized access attacks. This vulnerability becomes particularly concerning in challenging scenarios. Other research endeavors have designed data protection approaches in ICN-based wireless edge networks, categorizing them into authentication-based approaches [12], [13] and encryption-based solutions [14], [15]. Nonetheless, these approaches falter in capitalizing on ICN's data reuse capabilities and fall

short in furnishing comprehensive support for fine-grained access control over reused data within collaborative tasks.

In a pioneering investigation, Zhang et al. [16] introduced attribute-based encryption (ABE) to named data networking (NDN), which stands as a highly promising implementation of ICN [17]. Subsequently, several studies [18], [19], [20] have been dedicated to exploring the integration of ICN and ABE. Nevertheless, the current integration solutions of ICN and ABE cannot be directly applied to secure the reused data in task cooperation within ICN-based wireless edge networks, primarily due to the alterations in the communication service model. Concretely, current solutions primarily concentrate on the secure delivery of content from content producers to legitimate consumers who adhere to their respective access control policies, either directly or indirectly. In ICN-based task cooperation, the primary challenge lies in securely sharing data among cooperative devices to effectively execute tasks within the framework of a trusted control center, where a cooperative device can assume the roles of both a content producer and a consumer. Moreover, the current solutions fail to consider the existence of identical sub-policies within an access control policy. In the context of task cooperation, access control policies often share certain sub-policies (e.g., specific time range, specific location range, etc.). Optimizing the encryption process for collaborative task data with shared sub-policies can effectively reduce the security cost.

Based on the observations mentioned above, we present a security-enhanced task cooperation scheme for ICN-based challenging wireless edge networks, aiming to ensure the secure reuse of data. This paper inherits the foundational concept from our previous conference paper [21]. In this scheme, we introduce the ICN-based secure task cooperation model to quantify the cost associated with executing collaborative tasks. Additionally, we design a specific naming convention to facilitate collaborative task requests and the acquisition of keys pertaining to such tasks. Moreover, to mitigate security costs, we propose an enhanced name-based access control scheme that optimizes the encryption process for collaborative tasks sharing the same sub-policy, thereby reducing encryption cost. The key contributions of this paper can be succinctly summarized as follows.

- 1) We present a secure task cooperation model based on ICN in challenging wireless edge networks, which quantifies three types of costs associated with securely executing collaborative tasks: computation cost, security cost, and communication cost.
- 2) We extend the conventional naming convention to facilitate the retrieval of keys associated with collaborative tasks. This naming convention supports the sequential cooperative processing of subtasks that are derived from larger tasks. The keys utilized in collaborative tasks encompass content keys and attribute keys.
- 3) We propose an enhanced name-based access control scheme to ensure meticulous access control for data reuse in task cooperation. This policy incorporates time and location information pertaining to collaborative tasks, as well as the privileges assigned to collaborative devices. Furthermore, this mechanism optimizes the

encryption process for diverse access control policies that share the same sub-policy.

- 4) We implement and evaluate our proposed scheme on the ndnSIM platform. The experimental results demonstrate that our scheme effectively enhances the security of task cooperation, while incurring a relatively low additional security cost compared to existing schemes.

The structure of this paper is organized as follows. Section II provides a comprehensive review of the related work. In Section III, we present an example scenario to illustrate the context of our study. Section IV introduces the design goals, system model, and security assumptions. The design of our proposed scheme is presented in Section V. In Section VI, we conduct several experiments to evaluate the performance of our scheme. Security analysis and discussions are presented in Section VII. Finally, we conclude our work in Section VIII.

II. RELATED WORK

In this section, we provide a comprehensive review in ICN-based wireless edge networks, focusing on two main aspects: enhancing network resource utilization and strengthening network security.

A. Enhancing Network Resource Utilization

Lee et al. [8] proposed the concept of “compute reuse”, which involves leveraging the results of previously executed computational tasks, either partially or fully, to compute new results for multiple users. They also proposed an ICN-based computation reuse architecture for edge systems, which effectively implements this design philosophy. Through experimental studies, it was demonstrated that the proposed architecture has the potential to substantially reduce resource utilization and minimize the execution time required for new tasks. Tan et al. [7] presented an NDN-based distributed collaborative task scheduling scheme in mobile ad-hoc network scenarios, aiming to enhance the efficiency of the task execution. Tan’s scheme divides collaborative tasks into subtasks that can be effectively reused thanks to the in-network caching capabilities. Nour and Cherkaoui [9] developed a network-based computation reuse architecture that operates at both the networking and application levels. This architecture effectively stores and retrieves previously executed results to fulfill newly arrived tasks with similar requirements, thereby avoiding redundant computations. Bellal et al. [10] designed CoxNet, an efficient computation reuse architecture designed for edge computing. CoxNet enables edge servers to reuse previously computed results when scheduling incoming computations with dependencies. The performance of the architecture was evaluated using synthetic and real-world datasets. Furthermore, this paper presents an analytical model for computation reuse in conjunction with dependent task offloading and proposes a novel computing offloading scheduling scheme. Al Azad and Mastroakis [11] proposed Reservoir, a framework that facilitates pervasive computation reuse at the edge. Reservoir extends the NDN architecture and leverages Locality Sensitive Hashing (LSH) mechanisms to

enable computation reuse semantics directly within the edge network infrastructure.

However, the aforementioned mechanisms primarily concentrate on utilizing the computation reuse or collaborative task architecture to enhance resource utilization in edge networks, often overlooking the aspect of safeguarding against security threats associated with data reuse in collaborative tasks.

B. Strengthening Network Security

Oh et al. [12] proposed an extension to the content-centric networking architecture to accommodate military and emergency communications ad hoc network scenarios characterized by high mobility and a high packet loss rate. In their proposed architecture, the consumer needs to undergo authentication at the autonomous domain gateway in order to obtain the key required for decrypting the received data packet. Amadeo et al. [13] designed the protocol, named authenticated named data networking, which requires the content provider to dynamically maintain a public key list comprising authorized consumers. In previous studies [12], [13], the network entities responsible for authentication need to remain online, which may not be suitable for intermittently connected and challenging scenarios.

Misra et al. [14] proposed a data access control mechanism that utilizes threshold secret sharing. In their approach, the content producer can authorize legitimate consumers to decrypt data even if the entity responsible for authentication is off-line. However, in order to decrypt content from multiple content providers, consumers are required to store secret shares from each provider, resulting in additional cost that can be burdensome in resource-constrained challenging scenarios. Liu et al. [15] proposed a data access control mechanism that incorporates a pre-calculated and cached auxiliary key block to minimize the retrieval delay and decryption cost for consumers. Additionally, they integrated a two-dimensional one-way function into the mechanism to ensure the uniqueness of each consumer's secret share. Considering that the entities responsible for authentication no longer require continuous online presence, the aforementioned approaches [14], [15] become applicable in intermittently connected challenging scenarios. However, it is essential to acknowledge that these solutions do not fully meet the fine-grained access control requirements for data reuse in collaborative task scenarios. Furthermore, Kerberos, initially conceived at MIT, stands as a widely employed network authentication protocol primarily aimed at verifying user identities and issuing secure tokens to ensure resource access. Nonetheless, it lacks the capacity for enhanced flexibility in overseeing and regulating content sharing as well as access rights [22].

In 2018, Zhang et al. [16] firstly proposed a name-based access control mechanism in NDN, which presents the NAC-ABE library. Subsequently, several studies [18], [19], [20] have explored the integration of NDN and ABE. However, these schemes cannot be directly applied to secure reused data in task cooperation within ICN-based wireless edge networks, primarily due to changes in the communication service model.

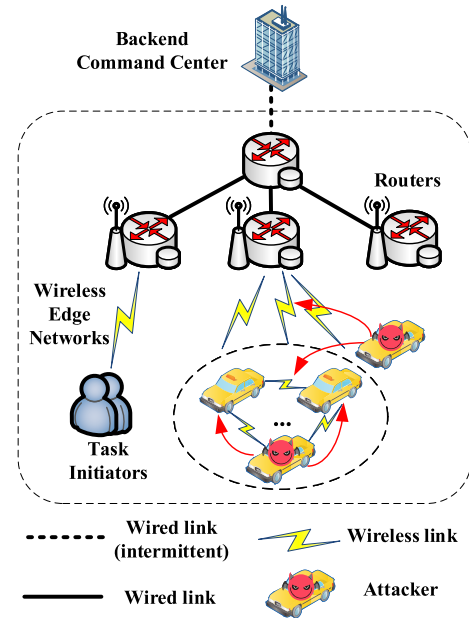


Fig. 1. Challenging disaster relief scenario.

Additionally, these schemes exhibit a higher encryption cost compared to the name-based access control mechanism when implementing the same sub-access control policy.

III. EXAMPLE SCENARIO

In Fig. 1, we illustrate a challenging wireless edge network environment that represents a typical disaster relief scenario. In this scenario, there are three main challenges and requirements as follows.

- 1) **Intermittent Connectivity:** The infrastructure in disaster-stricken areas may be damaged, leading to the inability to guarantee a continuous and stable connection path between the source and destination, traditional TCP/IP based client-server protocols are unable to support communication. On the other hand, ICN adopts a content-centric communication approach, allowing asynchronous content transmission between two nodes without the need for an end-to-end connection. This feature is considered suitable for such challenging scenarios.
- 2) **Limited Computational Resources:** Due to the complexity and diversity of rescue tasks, as well as the limited computational resources of individual devices, task cooperation is required. It is worth noting that the in-network caching feature of ICN allows for data reuse, which enhances the resource utilization in isolated networks.
- 3) **Vulnerability to Security Threats:** Disaster relief involves a large amount of sensitive information and critical tasks, making it highly susceptible to security threats such as eavesdropping and unauthorized access attacks. Therefore, on the one hand, during the task cooperation, collaborative data must be protected from being intercepted by malicious users, curious intermediate network devices. On the other hand, collaborative devices should

only be allowed to access collaborative data within their authorized access rights and should not have unauthorized access to other data. Also, the added security cost should be kept as low as possible. Furthermore, ICN adopts a content-centric security model, which inherently provides advantages in protecting content security.

IV. SYSTEM MODEL AND CRYPTOGRAPHIC PRELIMINARIES

In this section, we analyze the design goals of securing data reuse in ICN-based task cooperation. Subsequently, we present the system model and quantify the cost of securing data reuse. Finally, we provide a detailed explanation of the security assumptions and cryptographic preliminaries involved in the proposed scheme.

A. Design Goals

Building upon the aforementioned example scenarios, the proposed scheme need to achieve the following design goals.

- 1) *Fine-grained access control.* In the proposed scheme, the granularity of access control should be specific to the inherent properties of the task itself, such as time or location, or the authorization information of the entities executing the task, which ensures that only unmanned vehicles adhering to the access policy generated by the trusted control center are granted access to data reuse. It is worth noting that the granularity of access control also needs to align with the ICN's naming convention.
- 2) *Data reuse confidentiality.* The proposed scheme guarantees that unauthorized unmanned vehicles and cache-enabled routers are unable to obtain any information from data reuse.
- 3) *Lower computation and security cost.* The proposed scheme should be capable of leveraging the advantages of data reuse of ICN to reduce the computation cost of task cooperation. Additionally, considerations should be given to mitigating the security cost associated with fine-grained access control, such as through the appropriate reuse of access control policy.

B. System Model

We present the system model in Fig. 2, which includes various entities, including task initiators, attackers, unmanned vehicles, ICN routers, the trusted control center, and the attribute authority, among others.

- 1) *Task initiators* are responsible for reporting nearby emergencies to the trusted control center, providing information about the emergency type and current location. Once a response is received from the trusted control center, the task initiator proceeds to send task request packets accordingly.
- 2) *Attackers* have the ability to intercept packets transmitted or received by unmanned vehicles via wireless channels, allowing them to gain access to valuable information. Furthermore, attackers can manipulate these packets,

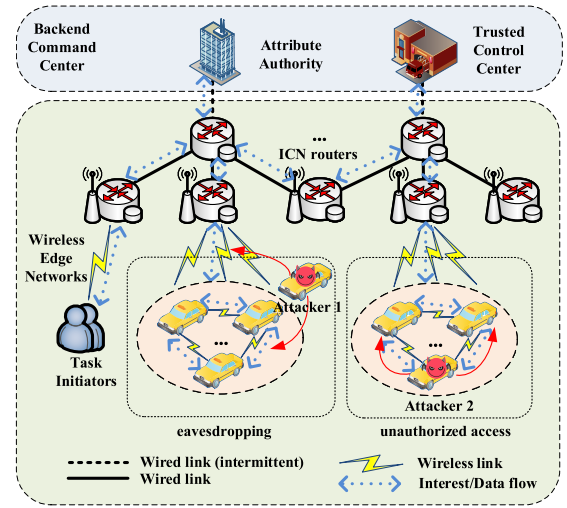


Fig. 2. System model.

potentially causing disruptions in rescue operations during disaster scenarios and compromising social order.

- 3) *Unmanned vehicles* play a vital role in performing collaborative tasks, which involve tasks such as real-time scenario detection and operational execution. Within the context of task cooperation, unmanned vehicles can act as both content producers and consumers. As they engage in data sharing, they are simultaneously exposed to various security threats, including eavesdropping attacks and unauthorized access attacks.
- 4) *ICN routers* encompass both wired and wireless types. These routers are responsible for forwarding requests and responding to them if the requested content is present in their cache. Furthermore, ICN routers can exhibit curiosity by accessing the contents cached within the routers that they are forwarding.
- 5) *Trusted control center* has the capability to select a combination of unmanned vehicles to carry out collaborative tasks. Moreover, the trusted control center generates attribute-based access control policies based on factors such as the current time, location of the collaborative tasks, and the privileges assigned to the participating unmanned vehicles.
- 6) *Attribute authority* is responsible for initializing the security parameters of the system and disseminating public parameters to the network. Furthermore, attribute authority generates attribute keys specifically for unmanned vehicles.

Regarding the physical network connectivity, wired ICN routers are linked through wired connections, while unmanned vehicles and wireless ICN routers are connected via wireless channels.

Then, we quantify the cost of securing data reuse in task cooperation. The common notations in this paper are summarized in Table I.

We make the assumption that there are n unmanned vehicles involved in the collaborative task, denoted as V_i ($1 \leq i \leq n$). Additionally, the collaborative task T can be partitioned into m subtasks, with T_k representing one of the subtasks executed

TABLE I
SUMMARY OF NOTATIONS

Notation	Description
V_i	The i -th unmanned vehicle
T_k	The k -th subtask executed sequentially in collaborative task T
c_k^i	Computation cost of V_i performing subtask T_k
τ_k^i	Flag indicating whether subtask T_k is reused in V_i
L_i	Lookup time of the data reuse table of V_i
Δ_k	Complexity of subtask T_k
φ_i	Computation capability of V_i
s_k^i	Security cost of V_i performing subtask T_k
I_k	Input data size of subtask T_k
O_k	Output data size of subtask T_k
α_i	Encryption capability coefficient of V_i
β_i	Decryption capability coefficient of V_i
d_k	Communication cost of executing subtask T_k
b_e	Bandwidth among unmanned vehicles
r_T	Encryption cost of the content key used in collaborative task T
a_T	Number of attributes of collaborative task T
a_s	Number of attributes of the same sub-policy
α_0	Encryption capability coefficient of the trusted control center

sequentially. The overall cost of performing the collaborative task comprises three components: 1) the computation cost associated with the task, 2) the security cost of ensuring data reuse protection, and 3) the communication cost among the unmanned vehicles.

Computation Cost: The computation cost of the task is determined by the computation capability of the unmanned vehicle executing each subtask and the complexity of the subtask. It encompasses the computation cost of each individual subtask. Moreover, if data reuse occurs within a subtask, the computation cost solely includes the lookup time of the data reuse table.

The computation cost c_k^i is defined as depicted in Equation (1), which denotes the computational expense incurred when unmanned vehicle V_i carries out subtask T_k .

$$c_k^i = L_i + \left(1 - \tau_k^i\right) \cdot \frac{\Delta_k}{\varphi_i} \quad (1)$$

where τ_k^i indicates whether subtask T_k is reused in V_i ($\tau_k^i = 1$) or not ($\tau_k^i = 0$), L_i represents the lookup time of V_i 's data reuse table, Δ_k corresponds to the complexity of subtask T_k , and φ_i denotes the computation capability of V_i .

Security Cost: The security cost of collaborative task execution encompasses the encryption cost of content keys used in collaborative tasks at the trusted control center and the encryption and decryption cost incurred by cooperative vehicles during the task execution phase. Specifically, the encryption cost of content keys is primarily influenced by the attribute granularity within the access control policy of the collaborative task. Furthermore, assuming a fixed length for the content key, the encryption and decryption cost associated with data reuse during subtask execution is determined by the encryption and decryption capabilities of unmanned vehicles, as well as the size of the input and output data for the subtask. Additionally, in cases where data reuse occurs, the encryption cost for the subtask's output data is zero.

The security cost at the trusted control center is defined by Equation (2), which signifies the encryption cost of the content key employed in collaborative task T .

$$r_T = (a_T - a_s) \cdot \alpha_0 \quad (2)$$

where a_T denotes the count of attributes associated with the collaborative task, a_s represents the number of attributes within the same sub-policy, and α_0 signifies the encryption capability coefficient of the trusted control center.

The security cost of cooperative vehicles, denoted as s_k^i , is defined by Equation (3). It quantifies the encryption and decryption cost incurred by unmanned vehicle V_i during the execution of subtask T_k .

$$s_k^i = \frac{I_k}{\beta_i \cdot \varphi_i} + \left(1 - \tau_k^i\right) \cdot \frac{O_k}{\alpha_i \cdot \varphi_i} \quad (3)$$

where I_k represents the input data size of subtask T_k , while O_k denotes the output data size of the same subtask. Additionally, α_i signifies the encryption capability coefficient of unmanned vehicle V_i , and β_i refers to its corresponding decryption capability coefficient.

Communication Cost: The communication cost between unmanned vehicles is determined by the size of the input and output data of the subtask, as well as the bandwidth available between the vehicles. In the case of NDN, which follows a "pull"-based communication paradigm, the communication cost comprises the transmission delay associated with Interest packets and Data packets.

The communication cost d_k is defined by Equation (4), representing the cost of communication during the execution of subtask T_k .

$$d_k = \frac{I_k + O_k}{b_e} \quad (4)$$

where b_e denotes the bandwidth between any two unmanned vehicles.

The objective function, represented as Eq. (5), aims to minimize the cost associated with collaborative task execution.

$$\min_{T_k \in T} r_T + \sum_{k=1}^m \left(d_k + s_k^i + c_k^i \right) \quad (5)$$

C. Security Assumptions

- 1) In challenging scenarios, collaborative tasks often pose greater complexity. Limited resources make it challenging for a single unmanned vehicle to accomplish complex tasks independently. We make the assumption that the task can be divided into subtasks, allowing for cooperative execution across multiple unmanned vehicles. Furthermore, these subtasks have a sequential relationship, where the output of one subtask becomes the input for the next.
- 2) In our scheme, the unmanned vehicles are considered potentially malicious. They exhibit behaviors such as eavesdropping, tampering, and attempting to destroy unauthorized data. Additionally, they may engage in collisions with one another or collaborate with cache-enabled routers to carry out the aforementioned attacks.

- 3) Cache-enabled routers play a crucial role in providing cached data for unmanned vehicles in collaborative tasks. We assume that these routers possess rational and curious characteristics. Rationally, they adhere to designated protocols that facilitate services for unmanned vehicles. Curiosity-wise, cache-enabled routers may exhibit an interest in the extensive information contained within the cached data. This information encompasses sensitive consumer privacy data and real-time detection information, particularly relevant in challenging scenarios.
- 4) The trusted control center and attribute authority serve as the management and control entities for collaborative tasks in our scheme. These entities are regarded as trustworthy.

D. Cryptographic Preliminaries

This paper presents a scheme that incorporates ciphertext-policy attribute-based encryption (CP-ABE) [23] utilizing bilinear pairings [24] based on the Decisional Bilinear Diffie-Hellman (DBDH) assumption [25] and a secret sharing scheme [26]. The proposed scheme is outlined as follows.

1) *Bilinear Pairing*: Consider $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ as multiplicative cyclic groups of the same prime order p . Let g_1 and g_2 denote the generators of $\mathbb{G}_1$ and $\mathbb{G}_2$ respectively. We define the map $\hat{e} : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ as a bilinear pairing [24], possessing the following properties.

- Bilinearity: $\forall g_1 \in \mathbb{G}_1, g_2 \in \mathbb{G}_2$ and $\forall a, b \in \mathbb{Z}_p$; $\hat{e}(g_1^a, g_2^b) = \hat{e}(g_1, g_2)^{ab}$.
- Non-degeneracy: $\exists g_1 \in \mathbb{G}_1, g_2 \in \mathbb{G}_2$; $\hat{e}(g_1, g_2) \neq 1_{\mathbb{G}_T}$.
- Computability: $\forall g_1 \in \mathbb{G}_1, g_2 \in \mathbb{G}_2$, there exists a polynomial algorithm to compute $\hat{e}(g_1, g_2)$.

2) *DBDH Assumption*: Consider $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ as multiplicative cyclic groups of the same prime order p . Let g_1 and g_2 denote the generators of $\mathbb{G}_1$ and $\mathbb{G}_2$ respectively. Given an input tuple $(g_1, g_2, g_2^a, g_2^b, g_2^c, Z)$, where $a, b, c \in \mathbb{Z}_p$ and $Z \in \mathbb{G}_T$, it becomes computationally challenging to determine whether Z equals $\hat{e}(g_1, g_2)^{abc}$. Formally, we define an adversary $\mathcal{A}$ to have a non-negligible advantage ϵ in solving the DBDH problem [25] if

$$\Pr \left[\mathcal{A}(g_1, g_2, g_2^a, g_2^b, g_2^c, \hat{e}(g_1, g_2)^{abc}) = 1 \right] - \Pr \left[\mathcal{A}(g_1, g_2, g_2^a, g_2^b, g_2^c, Z) = 1 \right] \geq \epsilon$$

3) *Secret Sharing Scheme*: The notion of secret sharing was first introduced by Shamir [26]. Shamir's threshold secret sharing scheme, a well-known classic scheme, is based on Lagrange polynomial interpolation and is presented as follows.

Let $\mathbb{Z}_q$ represent a finite field with q , and let $s \in \mathbb{Z}_q$ be the main secret to be shared. The initial step involves the dealer selecting a polynomial $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{k-1}x^{k-1}$ of degree $(k-1)$. This polynomial is chosen such that $f(0) = s$, where $a_0 = s$ and $a_i (i = 1, 2, \dots, k-1) \in \mathbb{Z}_q$. Subsequently, the dealer assigns n two-tuples $\{(x_i, f(x_i)) (i = 1, 2, \dots, n)\}$ to each player $P_i (i = 1, 2, \dots, n)$, where $x_i \in \mathbb{Z}_q$. Finally, the secret s is reconstructed by employing

Lagrange polynomial interpolation with k distinct two-tuples. It is important to note that n is significantly smaller than q , and $n \geq k$.

$$s = f(0) = \sum_{i=1}^k f(x_i) \lambda_i = \sum_{i=1}^k f(x_i) \prod_{1 \leq j \leq k, j \neq i} \frac{x_j}{x_j - x_i}$$

where λ_i represents the Lagrange coefficient. It has been empirically demonstrated that the retrieval of the secret s is attainable with at least k participants.

V. SCHEME DESIGN

In this section, we begin by establishing precise nomenclature for task requests and key-related packets in task cooperation, as a result of utilizing content names for addressing and routing. Additionally, we elaborate on the workflow of our mechanism and provide a comprehensive description of how our approach ensures data reuse with minimal expenditure.

A. Naming Convention

We have formulated a meticulous naming convention for the purpose of ensuring secure task cooperation. This convention is adhered to by all task request packets and key-related packets within our scheme.

1) *Task Request*: In the context of secure task cooperation, unmanned vehicles systematically execute subtasks in accordance with the predetermined execution order established by the trusted control center. During the task execution process, a task initiator is required to transmit a task request packet. The naming convention for the task request packet is precisely defined as follows.

$$\text{Task Request} = \langle \text{TCC prefix} \rangle / \langle \text{task name} \rangle / \langle \text{subtask order} \rangle / \langle \text{current subtask name} \rangle / \langle \text{parameter} \rangle$$

where the TCC denotes the entity responsible for collaborative task orchestration, the task name represents the specific task requiring execution, the subtask order signifies the prescribed sequence in which subtasks are performed, the current subtask name denotes the name of the subtask currently in execution, and the parameter denotes the input parameter associated with the ongoing subtask.

2) *Encryption Content Key (ECK)*: In order to ensure the encryption and decryption of data reuse in task cooperation, unmanned vehicles are required to retrieve the ECK packets, which are encrypted using a specific attribute policy. Subsequently, the unmanned vehicles decrypt the ECK packets to obtain the CK. The naming convention for the ECK packet is precisely defined as follows.

$$\begin{aligned} \text{ECK Interest} &= \langle \text{TCC prefix} \rangle / \langle \text{task name} \rangle / \text{ECK} \\ \text{ECK Data} &= \langle \text{TCC prefix} \rangle / \langle \text{task name} \rangle / \text{ECK} \\ &\quad / \text{ENCRYPTED-BY} / \langle \text{attribute policy} \rangle \end{aligned}$$

where the TCC serves as the entity responsible for generating and encrypting the CK, the task name refers to the specific name assigned to the task being executed by the cooperative

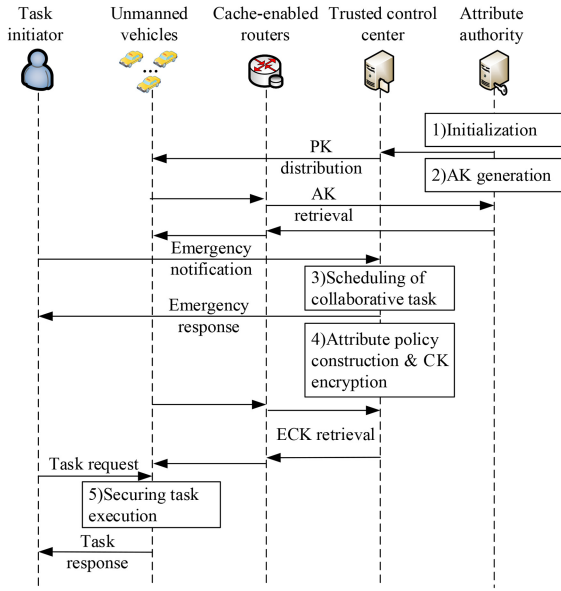


Fig. 3. Workflow of securing task cooperation.

unmanned vehicle, and the attribute policy determines the level of granularity associated with data reuse within the task.

3) *Attribute Key (AK)*: In order to decrypt the ECK packet, unmanned vehicles are required to obtain the AK packets from the attribute authority. The naming convention for the AK packet is precisely defined as follows.

$$\begin{aligned} \text{AK Interest} &= \langle \text{AA prefix} \rangle / \langle \text{attribute name} \rangle \\ &\quad / \text{ENCRYPTED-BY} / \langle \text{unmanned-vehicle-private-key-id} \rangle \\ \text{AK Data} &= \langle \text{AA prefix} \rangle / \langle \text{attribute name} \rangle \\ &\quad / \text{ENCRYPTED-BY} / \langle \text{unmanned-vehicle-public-key-id} \rangle \end{aligned}$$

where the AA is the entity responsible for generating the AK, the attribute name represents the attribute set associated with the unmanned vehicle, and the private key ID and public key ID refer to a pair of keys (public and private) unique to the unmanned vehicle.

B. Workflow

The workflow of our scheme encompasses five distinct steps, as depicted in Fig. 3. The initial two steps are carried out at the attribute authority, while the subsequent two steps are executed at the trusted control center. The final step necessitates collaborative execution among the unmanned vehicles.

1) *Initialization*: In this step, the attribute authority begins by generating the public key (PK) and the master secret key (MSK). To accomplish this, the attribute authority selects a bilinear pairing function $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, along with random exponents α and $\beta \in \mathbb{Z}_q$, where $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ represent multiplicative cyclic groups of a large prime order p . Let g_1 and g_2 denote the generators of $\mathbb{G}_1$ and $\mathbb{G}_2$ respectively. Additionally, the attribute authority selects a collision-resistant hash function $H_1 : \mathbb{G}_T \rightarrow \{0, 1\}^n$ and $H_2 : \{0, 1\}^* \rightarrow \mathbb{G}_1$. Subsequently, the attribute authority generates PK and MSK according to the following

procedure.

$$\begin{aligned} \text{PK} &= \{ \mathbb{G}_1, \mathbb{G}_2, g_1, g_2, g_1^\beta, e(g_1, g_2)^\alpha \}, \\ \text{MSK} &= \{ \beta, g_2^\alpha \}. \end{aligned} \quad (6)$$

Finally, the PK needs to be distributed to both the trusted control center and the unmanned vehicles that are involved in the task cooperation process.

2) *AK Generation*: In this step, the attribute authority is tasked with generating the respective AKs for each V_i , based on their attribute sets. To accomplish this, the attribute authority initiates the process by selecting a random parameter $r \in \mathbb{Z}_p$ and assigns $r_i \in \mathbb{Z}_p$ to each attribute $i \in \mathcal{B}$, where $\mathcal{B}$ represents the attribute sets of all participating unmanned vehicles in the cooperative task. Subsequently, the AK can be generated using the following procedure.

$$\begin{aligned} \text{AK}_i &= \left\{ D = g_2^{\frac{\alpha+r}{\beta}}, D_i = g_2^r \cdot H_2(i)^{r_i}, \right. \\ &\quad \left. D'_i = g_1^{r_i} \right\}, i \in \mathcal{B}. \\ \text{AK} &= \{ \text{AK}_i, \forall i \in \mathcal{B} \}. \end{aligned} \quad (7)$$

During the execution of cooperative tasks, each V_i is required to periodically send an Interest packet following the specific naming convention (as described in Section V-A3), in order to obtain the attribute key. This is achieved by utilizing its unique public-private key pair.

3) *Scheduling of Collaborative Task*: In the event of an emergency requiring the collective efforts of unmanned vehicles, the task initiator will transmit an emergency notification packet to the trusted control center. Upon receipt of the notification packet, the trusted control center will carefully select a group of suitable devices for executing the emergency tasks, taking into account their current locations and computational capabilities. Furthermore, the trusted control center is responsible for conveying the scheduled task execution sequence to the task initiator. This information enables the task initiator to construct a task request packet adhering to the specific naming convention (in Section V-A1).

4) *Attribute Policy Construction & CK Encryption*: During this step, the trusted control center is responsible for generating a tailored access control policy. This policy is then utilized to encrypt the content key, which in turn is used for encrypting the shared data in task cooperation. By encrypting the shared data, only authorized unmanned vehicles that adhere to the specific access control policy can access and utilize the data within the cooperative task. Moreover, given the resource limitations of terminals in challenging scenarios, our mechanism supports the reuse of a portion of the encryption process across multiple collaborative tasks that share the same sub-policy. This approach effectively reduces the overall security cost.

To illustrate the generation process of the access control policy, we will use two collaborative tasks with the same sub-policy as an example. This example can be easily extended to multiple cooperative tasks sharing the same sub-policy. As depicted in Fig. 4, the access control policies Γ_1 and Γ_2 correspond to the two distinct collaborative tasks. However, both policies share the same sub-policy, denoted as Γ . This

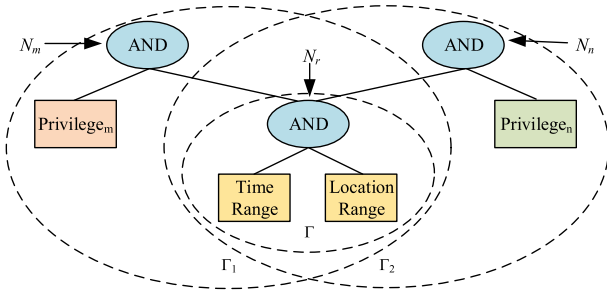


Fig. 4. Two name-based access control policies with the same sub-policy.

sub-policy encompasses time and location information pertaining to the collaborative tasks, as well as the privileges assigned to the unmanned vehicles involved. It is important to note that the trusted control center is responsible for constructing the access control policy. In this context, we designate the root nodes of Γ_1 , Γ_2 , and Γ as N_m , N_n , and N_r respectively.

Subsequently, the trusted control center proceeds to generate polynomials $f_t(x)$ for each node N_t in both Γ_1 and Γ_2 . Here, N_t represents any given node within Γ_1 and Γ_2 . The degree of polynomial $f_t(x)$ is determined by the threshold value of the corresponding node, denoted as k_t . Specifically, the degree d_t of $f_t(x)$ is calculated as $d_t = k_t - 1$. With node N_r as the starting point, the polynomial for each node N_t is generated using the following process. A random number $s \in \mathbb{Z}_p$ is selected, and $f_r(0)$ is assigned this value. Additionally, $f_r(x)$ is computed by randomly selecting d_r other points. For the parent nodes N_n and N_m of N_r , Equation (8) holds.

$$f_r(0) = f_m(\text{index}(N_r)) = f_n(\text{index}(N_r)) \quad (8)$$

To obtain the values of m_0 and n_0 , we set $f_m(x) = m_{d_m}x^{d_m} + m_{d_m-1}x^{d_m-1} + \dots + m_1x + m_0$ and $f_n(x) = n_{d_n}x^{d_n} + n_{d_n-1}x^{d_n-1} + \dots + n_1x + n_0$, with $x = 1$. Equation (8) yields $s = f_m(1) = f_n(1)$, which implies $s = m_{d_m} + m_{d_m-1} + \dots + m_0 = n_{d_n} + n_{d_n-1} + \dots + n_0$. By selecting $d_m + d_n$ random coefficients for the polynomials, excluding m_0 and n_0 , and applying Equation (9) and Equation (10), the values of m_0 and n_0 can be calculated. Ultimately, the trusted control center determines the polynomials $f_m(x)$ and $f_n(x)$.

$$m_0 = s - (m_{d_m} + m_{d_m-1} + \dots + m_1) \quad (9)$$

$$n_0 = s - (n_{d_n} + n_{d_n-1} + \dots + n_1) \quad (10)$$

For each node N_t in Γ_1 and Γ_2 , excluding the nodes N_n , N_m , and N_r , we assign $f_t(0) = f_{\text{parent}(N_t)}(\text{index}(N_t))$. Additional d_t points are selected to determine $f_t(x)$ for each node N_t . This process allows us to determine the polynomial for each individual node. Let $\mathcal{Y}_1$ and $\mathcal{Y}_2$ represent the sets of leaf nodes in Γ_1 and Γ_2 , respectively. Consequently, the trusted control center calculates and generates the ciphertexts M_1 and M_2 for the content keys CK_1 and CK_2 , respectively, using Equations (11) and (12).

$$\begin{aligned} M_1 = \{ & \Gamma_1, \tilde{C} = CK_1 \cdot e(g_1, g_2)^{\alpha f_m(0)}, C = g_1^{\beta f_m(0)}, \\ & C_{N_y} = g_1^{f_{N_y}(0)}, C'_{N_y} = H_2(\text{att}(N_y))^{f_y(0)}, \\ & \forall N_y \in \mathcal{Y}_1 \} \end{aligned} \quad (11)$$

$$\begin{aligned} M_2 = \{ & \Gamma_2, \tilde{C} = CK_2 \cdot e(g_1, g_2)^{\alpha f_n(0)}, C = g_1^{\beta f_n(0)}, \\ & C_{N_y} = g_1^{f_{N_y}(0)}, C'_{N_y} = H_2(\text{att}(N_y))^{f_y(0)}, \\ & \forall N_y \in \mathcal{Y}_2 \} \end{aligned} \quad (12)$$

During the encryption process of the content key, for the leaf nodes within the same sub-policy Γ , a single calculation is sufficient for both Equation (11) and Equation (12). Subsequently, the trusted control center generates CK packets in accordance with the naming convention and distributes them throughout the network.

5) *Securing Data Reuse in Task Execution*: In this step, the cooperative unmanned vehicles perform the collaborative tasks securely in accordance with the predetermined sequence provided by the trusted control center. Specifically, each cooperative unmanned vehicle decrypts the ECK using its respective AK, thereby obtaining the corresponding content key. With the acquired keys, the cooperative unmanned vehicles execute the corresponding subtask and utilize the content key to encrypt the intermediate data. This encryption ensures the confidentiality of the data reuse within the task cooperation.

As illustrated in Fig. 5, we present an illustration using three unmanned vehicles to demonstrate the secure execution of collaborative tasks in our proposed scheme. Let us consider the collaborative task T , which is divided into three subtasks: T_1 , T_2 , and T_3 , each executed by unmanned vehicles V_1 , V_2 , and V_3 respectively. Additionally, we assume that the attribute set of each unmanned vehicle is denoted as $\mathcal{B}$, and the attribute key corresponding to V_i is denoted as AK_i . Legitimate unmanned vehicles can obtain the CK by individually decrypting the encrypted ECK using their respective AK_i . The decryption process of the content key involves a recursive step, which is described in detail in Section VII-A, where the decryption proof is presented.

Let N_j represent a node in the attribute policy Γ_1 . If N_j is a leaf node within Γ_1 and $i = \text{att}(N_j) \in \mathcal{B}$, then the boolean value of N_j is set to true. The corresponding value of F_j can be computed using Equation (13). On the other hand, if N_j is a leaf node within Γ_1 and $i = \text{att}(N_j) \notin \mathcal{B}$, then the boolean value of N_j is set to false.

$$\begin{aligned} F_j = & \text{DecryptNode}(M_1, AK, N_j) \\ = & \frac{e(C_{N_j}, D_i)}{e(C'_{N_j}, D'_i)} \\ = & e(g_1, g_2)^{r f_j(0)} \end{aligned} \quad (13)$$

If the node N_j is not a leaf node in Γ_1 , we define the set of all leaf nodes of N_j as N_c . Let S_j represent a set consisting of $k_j \cdot N_c$. For each leaf node N_c within the set S_j , the boolean value is set to true. If the set S_j does not exist, the boolean value of N_j is false. Conversely, if the set S_j exists, the boolean value of N_j is true, and the value of F_j can be computed using

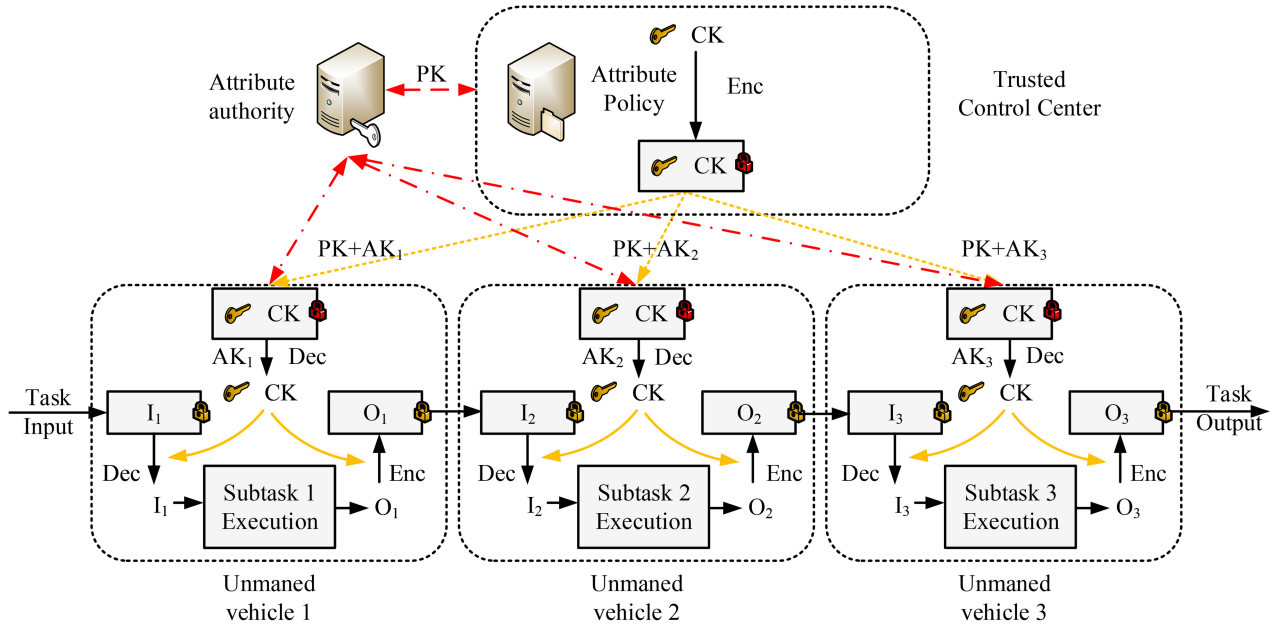


Fig. 5. Secure data reuse in unmanned vehicles.

Equation (14).

$$F_j = \prod_{z \in S_j} F_z^{\Delta_{z,S_j}(0)} = e(g_1, g_2)^{r \cdot f_j(0)} \quad (14)$$

where $\Delta_{z,S_j}(x) = \prod_{u \in S_j, u \neq v} \frac{x-u}{v-u}$.

When the attribute set of unmanned vehicles, denoted as $\mathcal{B}$, satisfies the access control policy Γ_1 , the value of F_m at the root node N_m is computed using Equation (15).

$$\begin{aligned} F_m &= \text{DecryptNode}(M_1, AK, N_m) \\ &= \frac{e(C_{N_m}, D_i)}{e(C'_{N_m}, D'_i)} \\ &= e(g_1, g_2)^{r f_m(0)} \end{aligned} \quad (15)$$

Lastly, the content key CK_1 is obtained by evaluating Equation (16). The decryption process for obtaining CK_2 from M_2 follows a similar set of steps as described above.

$$\frac{\tilde{C} \cdot F_m}{e(C, D)} = CK_1 \quad (16)$$

Once the cooperative unmanned vehicles have obtained the content keys, they can safely and collaboratively execute the collaborative task initiated by the task initiator. The task initiator constructs the task request packet based on the arrangement made by the trusted control center. In this packet, the input data of the task is encrypted using CK_1 , which can be the encryption key in symmetric encryption algorithms, such as AES-128 or AES-256. After constructing the data packet, the task initiator sends the task request packet, and subsequently, V_1 , V_2 , and V_3 sequentially execute the cooperative task K . Upon receiving the task request packet from the task initiator, V_1 proceeds to execute subtask T_1 following the processing procedure outlined in Algorithm 1. This process aligns with the subtask processing procedure followed by other unmanned vehicles participating in the cooperative endeavor.

Algorithm 1 The Processing Process of Subtask in the V_i

Input:

The k -th subtask executed in sequentially in task T , T_k ;
The input data size of T_k , I_k ;

Output:

The output data size of T_k , O_k ;

```

1: if  $T_k \in T$  then
2:   Input = Decrypt( $I_k$ )
3:   Prefix = Input.subtaskprefix()
4:   Para = Input.parameter()
5:   Flag = SubtaskReuseTable.query(Prefix, Para)
6:   if Flag  $\neq \emptyset$  then
7:      $O_k$  = SubtaskReuseTable.output()
8:   else
9:     Output =  $T_k$ .compute(Para)
10:     $O_k$  = Encrypt(Output)
11:    SubtaskReuseTable.add(Prefix, Para,  $O_k$ )
12:   end if
13:   return  $O_k$ 
14:   Forward  $T_k$  to other unmanned vehicles
15: end if

```

Initially, V_1 decrypts the input I_k using CK_1 . Subsequently, V_1 conducts a search within local data reuse table based on the subtask prefix and input parameters. If a matching entry is discovered in the table, the corresponding output result is retrieved. In cases where no match is found, V_1 proceeds to carry out the subtask T_k . In the final step, the output result is encrypted using CK_1 and returned to the unmanned vehicle that performs the subsequent subtask. Simultaneously, the task request for T is transmitted to other cooperative unmanned vehicles, facilitating the successive execution of the remaining subtasks.

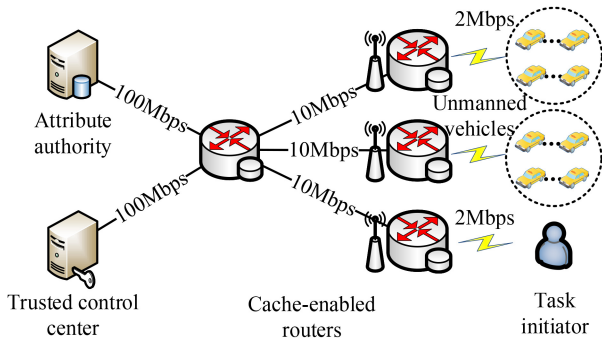


Fig. 6. Simulation topology.

TABLE II
EXPERIMENT PARAMETERS

Parameter	Value
Number of unmanned vehicles	[5-50]
Number of collaborative tasks	[5-50]
Subtask execution time	1s
Cache capacity of each NDN router	100K
Replacement policy	LRU

VI. PERFORMANCE EVALUATION

In this section, we begin by implementing our proposed scheme using the open-source network simulation platform ndnSIM. Subsequently, we proceed to evaluate the performance of our scheme using four key metrics.

A. Experiment Environment and Parameters

Utilizing the NAC-ABE library [16], we have successfully implemented the proposed ICN-based secure task cooperation scheme. The implementation of our scheme is integrated with the existing NDN using ndnSIM [27] for simulation purposes. To demonstrate the feasibility of our scheme, we constructed a proof of concept involving five entities: attribute authority, trusted control center, task initiator, NDN routers, and unmanned vehicles. The task initiator initiates the process by sending an emergency notification to the trusted control center. Upon receiving the notification, the trusted control center selects a suitable group of unmanned vehicles and coordinates the execution of collaborative tasks. Subsequently, the trusted control center provides instructions to the task initiator for constructing the task request packet. Finally, with the trusted control center orchestrating the process, the unmanned vehicles engage in collaborative task execution.

The simulation topology is depicted in Fig. 6. Recognizing that the trusted control center and attribute authority are typically situated at the backend, we have configured the transmission delay and link bandwidth of the connection between the trusted control center (or attribute authority) and the wireless edge network to 100ms and 100Mbps, respectively. Moreover, the transmission delay and link bandwidth between NDN routers have been set to 10ms and 10Mbps, respectively.

Table II presents a comprehensive overview of the experimental parameters. During the simulation, task initiators generate a total of 50 collaborative tasks, with each task comprising three subtasks. As our primary focus lies in the

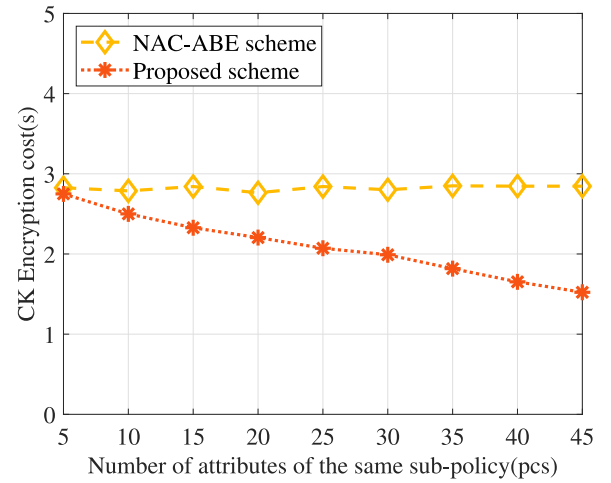


Fig. 7. The influence on the CK encryption cost of the number of attributes in the same sub-policy.

security cost introduced by the proposed scheme, we assume a uniform execution time of 1s for all subtasks. Additionally, the content key (CK) has a length of 256 bits, and the input (I_k) and output (O_k) sizes for all subtasks are fixed at 1KB (KiloBytes).

The experiments were conducted on a laptop equipped with an Intel Core i7-3770 CPU (4 cores, 3.40 GHz/core), 16GB RAM, and running the Ubuntu 16.04 operating system.

B. Experimental Results and Analysis

We assess the performance of the proposed scheme based on four key metrics: (1) encryption cost, (2) average retrieval delay of keys, (3) subtask reuse rate, and (4) total execution time of collaborative tasks. All experimental results presented in this study represent the average values obtained from 1000 independent runs.

1) *Encryption Cost*: The encryption cost in the proposed scheme encompasses the security initialization cost, the CK encryption cost, and the content encryption cost. We have optimized the process of CK encryption with the same sub-policy, resulting in improved efficiency compared to the NAC-ABE scheme.

Fig. 7 illustrates the impact of the number of attributes within the same sub-policy on the encryption cost of the CK, while keeping the number of CKs (fixed at 2) and the number of attributes in the access control policy (fixed at 50) constant. Then, we use the term “access control policy” to refer to the whole access control policy. In the proposed scheme, the encryption cost of the CK decreases as the number of attributes within the sub-policy increases. Conversely, in the NAC-ABE scheme, the encryption cost of the CK remains relatively constant. These findings demonstrate that the proposed scheme exhibits lower CK encryption costs in scenarios involving cooperative tasks with the same sub-policy.

Fig. 8 depicts the impact of the number of attributes in the access control policy on the encryption cost of the CK, while maintaining a constant ratio between the number of CKs (fixed at 2) and the number of attributes within the same sub-policy,

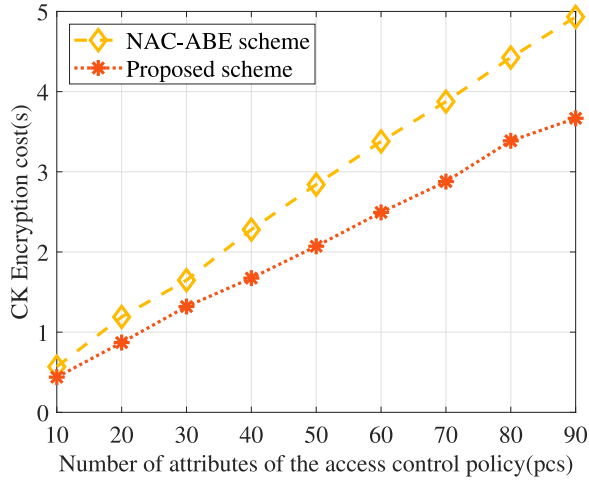


Fig. 8. The influence on the CK encryption cost of the number of attributes in the access control policy.

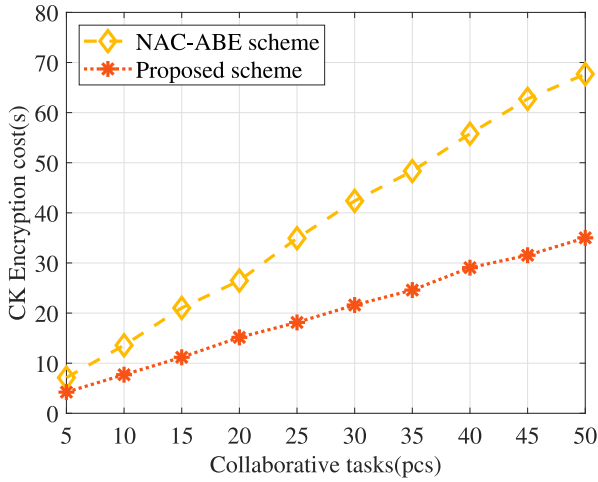
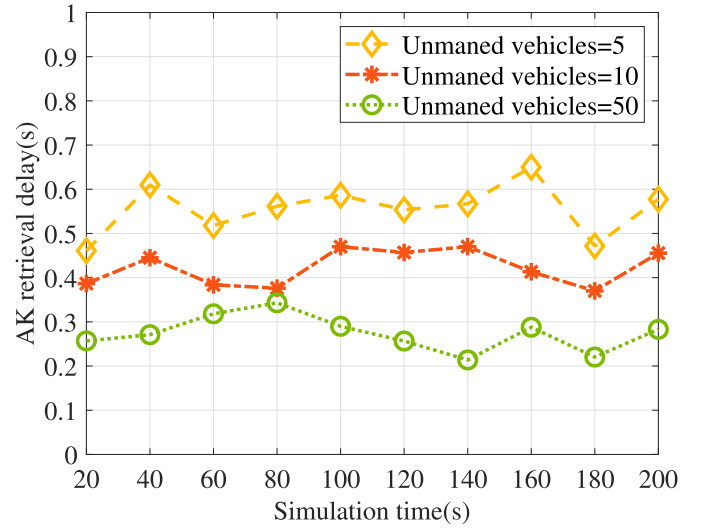


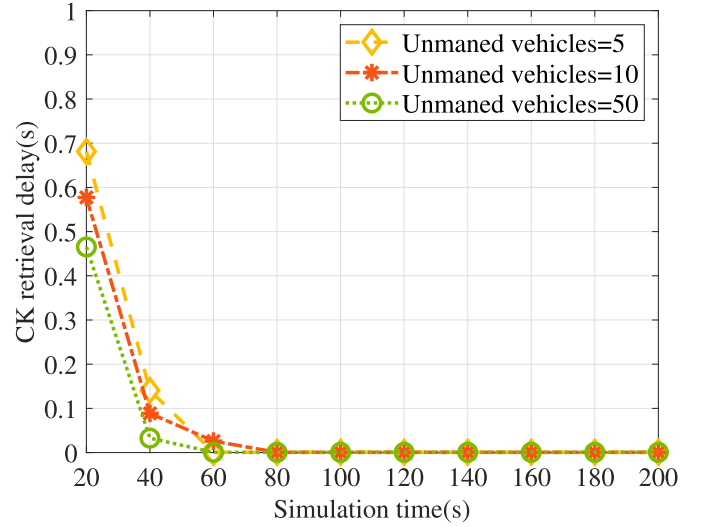
Fig. 9. The influence on the CK encryption cost of the number of collaborative tasks.

relative to the total number of attributes (set at 50%). Both the proposed scheme and the NAC-ABE scheme exhibit a positive correlation between the encryption cost of the CK and the number of attributes in the access control policy. However, the growth rate of encryption cost in the NAC-ABE scheme is higher than that in the proposed scheme. These findings indicate that the proposed scheme offers improved encryption efficiency in scenarios involving cooperative tasks with the same sub-policy.

Fig. 9 demonstrates the impact of the number of collaborative tasks on the encryption cost, while keeping the number of attributes within the same sub-policy (fixed at 25) and the number of attributes in the access control policy (fixed at 50) constant. Both the proposed scheme and the NAC-ABE scheme exhibit a positive correlation between the encryption cost of the CK and the number of collaborative tasks. However, the proposed scheme demonstrates a lower growth rate in encryption cost. Consequently, when the number of collaborative tasks increases, the proposed scheme exhibits higher encryption efficiency.



(a) Attribute Key.



(b) Content Key.

Fig. 10. Key retrieval delay.

2) *Average Retrieval Delay of Keys*: To ensure the security of data reuse in task cooperation, we have introduced two types of keys: AK and CK. Fig. 10 illustrates the key retrieval delay experienced by unmanned vehicles when acquiring these two types of keys, considering varying numbers of unmanned vehicles.

Fig. 10(a) illustrates that the average retrieval delay of AK remains relatively stable as the simulation time increases. This is because each unmanned vehicle possesses a specific AK, and the in-network caching has minimal impact on the retrieval delay of AK. Furthermore, the retrieval delay of AK decreases as the number of unmanned vehicles increases. This can be attributed to the higher vehicle density, resulting in more frequent connections between unmanned vehicles and a reduced need for data packet retransmissions. However, it is worth noting that the retrieval delay of AK has only a slight impact on our scheme, as AK is typically obtained at the beginning of task cooperation.

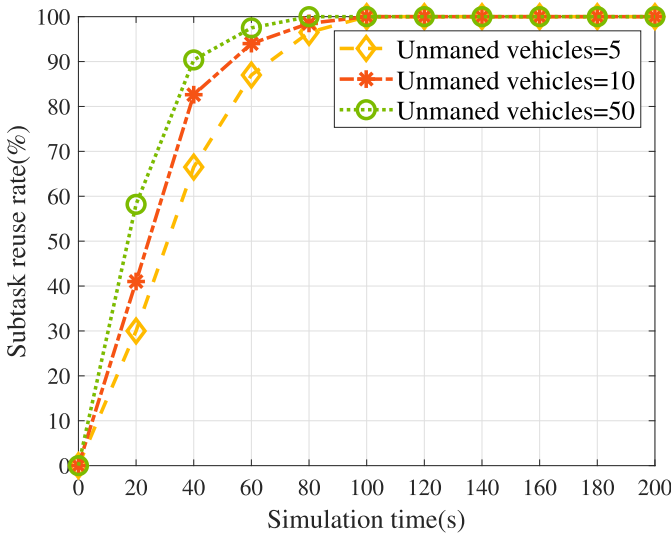


Fig. 11. Subtask reuse rate.

Fig. 10(b) presents the average retrieval delay of CK. It can be observed that the retrieval delay of CK decreases rapidly with the increase in simulation time across different numbers of unmanned vehicles. This reduction in retrieval delay is attributed to the impact of in-network caching, which effectively reduces the time required for unmanned vehicles to obtain CK. Consequently, our scheme introduces a low cost for unmanned vehicles in terms of retrieving the two types of keys.

3) *Subtask Reuse Rate*: Fig. 11 illustrates the subtask reuse rate across different numbers of unmanned vehicles. The subtask reuse rate represents the proportion of reused subtask results to the total number of subtasks. It can be observed that the subtask reuse rate increases in relation to the simulation time. Moreover, for the same duration, a higher number of unmanned vehicles leads to a higher subtask reuse rate. This indicates that an increased number of unmanned vehicles enhances the efficiency of task execution by facilitating the reuse of subtask results.

4) *Total Task Execution Time*: Fig. 12 presents a comparative analysis of the traditional scheme, the Tan et al. scheme [7], the dummy scheme, the NAC-ABE scheme [16], and the proposed scheme in terms of the total execution time of task cooperation. In the experiment, we conducted a total of 50 collaborative tasks and varied the number of unmanned vehicles participating in task cooperation from 5 to 50.

- *Traditional scheme*. The scheme does not factor in the reuse of subtask data and does not furnish any protective measures for the task data.
- *Tan et al. scheme*. The scheme solely takes into account the reuse of subtask data without introducing supplementary protective measures for the task data.
- *Dummy scheme*. The scheme takes into account the reuse of subtask data and offers safeguarding for repurposed data through the utilization of RSA and AES encryption algorithms.
- *NAC-ABE scheme*. The scheme takes into consideration the reuse of subtask data and ensures the security of

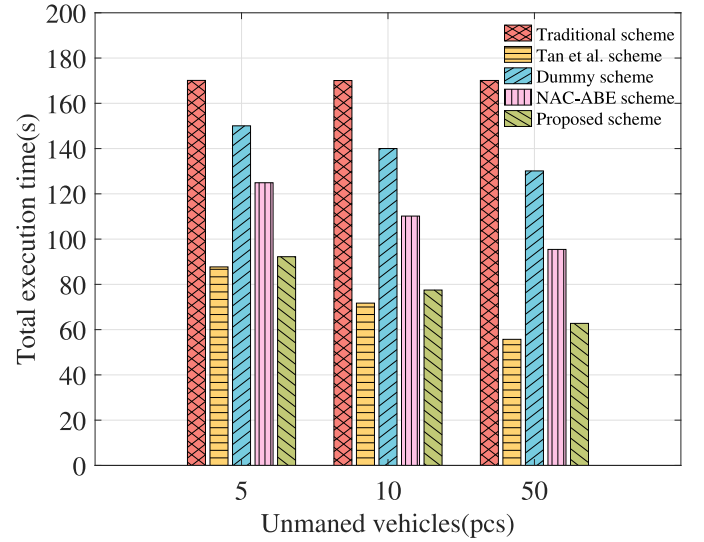


Fig. 12. Total execution time.

recycled data through the conventional CP-ABE and AES encryption algorithms.

- *Proposed scheme*. The scheme takes into account subtask data reuse and safeguards the repurposed data through an enhanced combination of CP-ABE and AES encryption algorithms.

The outcomes underscore the effectiveness of the proposed scheme in curtailing overall execution time when contrasted with traditional schemes, yielding a noteworthy reduction of 49.5% thanks to subtask reuse. Furthermore, the proposed scheme surpasses the dummy scheme, realizing a 44.7% time saving stemming from the diminished cryptographic operations. In comparison to the NAC-ABE scheme, the proposed approach achieves a substantial 29.7% time reduction due to the lowered encryption expenditure for collaborative tasks governed by the same sub-policy. It's crucial to acknowledge that the proposed scheme introduces a marginal time increment associated with security measures, resulting in an 8% uptick in total execution time compared to the Tan et al. scheme.

VII. SECURITY ANALYSIS AND DISCUSSION

In this section, we begin by validating the correctness of the proposed scheme, followed by a comprehensive analysis of its security. We demonstrate how our scheme effectively addresses potential threats such as eavesdropping attacks and unauthorized access attacks, ensuring robust protection of sensitive data. Additionally, we assess the cryptographic operations employed in the proposed scheme to evaluate its efficiency and effectiveness.

A. Correctness Validation

When the attribute set $\mathcal{B}$ of the cooperative vehicle complies with the access control policy Γ_1 , the decryption algorithm is initiated from the root node N_m of the access control policy Γ_1 . For the leaf nodes in Γ_1 , the calculation of F_j is performed

TABLE III
COMPARISON OF CRYPTOGRAPHIC OPERATIONS

Entity Type	Dummy Scheme (with RSA)	NAC-ABE Scheme (with CP-ABE)	Proposed Scheme (with enhanced CP-ABE)
Trusted Control Center (or Attribute Authority)	RSA KeyGen: M RSA Encryption: $M \times L$ RSA Decryption: $M \times L$	CP-ABE KeyGen: $a+2b$ CP-ABE Encryption: $(a+2b) \times L$ CP-ABE Decryption: $M \times L$	enhanced CP-ABE KeyGen: $a+2b$ enhanced CP-ABE Encryption: $(a+b) \times L^1$ enhanced CP-ABE Decryption: $M \times L$
Unmanned vehicles	AES Encryption: $M \times N$ AES Decryption: $M \times N$	AES Encryption: $\leq M \times N$ AES Decryption: $\leq M \times N$	AES Encryption: $\leq M \times N^2$ AES Decryption: $\leq M \times N^2$

¹ $(a+b) \times L$ can be enhanced by leveraging the reuse of the encryption process when employing the same sub-policy in the enhanced CP-ABE.

² $M \times N$ can be significantly reduced due to the efficient data reuse during task cooperation.

based on Equation (17).

$$\begin{aligned}
 F_j &= \frac{e(C_{N_j}, D_i)}{e(C'_{N_j}, D'_i)} = \frac{e(g_1^{f_j(0)}, g_2^r \cdot H_2(i)^{r_i})}{e(H_2(att(N_j))^{f_j(0)}, g_1^{r_i})} \\
 &= \frac{e(g_1^{f_j(0)}, g_2^r) \cdot e(g_1^{f_j(0)}, H_2(i)^{r_i})}{e(H_2(i)^{f_j(0)}, g_1^{r_i})} \\
 &= \frac{e(g_1, g_2)^{rf_j(0)} \cdot e(g_1, H_2(i))^{r_i f_j(0)}}{e(g_1, H_2(i))^{r_i f_j(0)}} \\
 &= e(g_1, g_2)^{rf_j(0)} \quad (17)
 \end{aligned}$$

For the non-leaf nodes in Γ_1 , the calculation of F_j is performed based on Equation (18).

$$\begin{aligned}
 F_j &= \prod_{z \in S_j} F_z^{\Delta_{z, S_j}(0)} \\
 &= \prod_{z \in S_j} (e(g_1, g_2)^{rf_z(0)})^{\Delta_{z, S_j}(0)} \\
 &= \prod_{z \in S_j} (e(g_1, g_2)^{rf_j(z)})^{\Delta_{z, S_j}(0)} \\
 &= e(g_1, g_2)^{r \sum_{z \in S_j} f_j(z) \Delta_{z, S_j}(0)} = e(g_1, g_2)^{rf_j(0)} \quad (18)
 \end{aligned}$$

Finally, by utilizing the bilinear property, the calculation of Equation (19) is performed, resulting in the successful recovery of CK_1 . This demonstrates the correctness of the proposed scheme.

$$\begin{aligned}
 \frac{\tilde{C} \cdot F_m}{e(C, D)} &= \frac{CK_1 \cdot e(g_1, g_2)^{\alpha f_m(0)} \cdot e(g_1, g_2)^{rf_m(0)}}{e(g_1^{\beta f_m(0)}, g_2^{(\alpha+r)/\beta})} \\
 &= \frac{CK_1 \cdot e(g_1, g_2)^{\alpha f_m(0) + rf_m(0)}}{e(g_1, g_2)^{\beta f_m(0) \cdot (\alpha+r)/\beta}} \\
 &= \frac{CK_1 \cdot e(g_1, g_2)^{(\alpha+r)f_m(0)}}{e(g_1, g_2)^{(\alpha+r)f_m(0)}} = CK_1 \quad (19)
 \end{aligned}$$

B. Data Confidentiality

During eavesdropping attacks, potential attackers can intercept data packets of collaborative tasks either by sniffing on the broadcast channel or obtaining them from in-network caching. However, it is important to note that these attackers are unable to decipher the meaning of the ciphertext. This is due to the encryption of all sensitive content, including the input and

output data of subtasks, as well as the CK. As a result, the confidentiality of the information remains preserved.

C. Fine-Grained Access Control

In unauthorized access attacks, legitimate unmanned vehicles may attempt to access data beyond their granted privileges. However, our scheme incorporates measures to address this issue by implementing fine-grained access control through the trusted control center. By restricting the granularity of authorized unmanned vehicles, the scheme ensures controlled access to the data.

For instance, consider the attribute policy assigned to the ECK Data: “/emergency-control /detection /ECK /ENCRYPTED-BY /(time₁ AND location₁) AND privilege₁ AND privilege₂”. By employing this distinct attribute policy, any data packet generated within the ECK framework remains impervious to decryption by unmanned vehicles situated at location₂ or by entities lacking the designated privileges, such as privilege₃.

This attribute-based policy provides a mechanism for precise access control, ensuring that only authorized unmanned vehicles meeting the defined conditions can decrypt and access the data.

D. Security Cost Analysis

We conducted a comparative evaluation of the cryptographic operations employed in the proposed scheme, in comparison to the existing dummy scheme and NAC-ABE scheme. Our evaluation considered L unmanned vehicles and M collaborative tasks, each comprising multiple subtasks with varying granularities, totaling N subtasks. Excluding data reuse, the total number of subtasks to be executed is $M \times N$.

In the dummy scheme, implemented with the RSA algorithm, each unmanned vehicle was granted access rights to all data associated with the $M \times N$ subtasks. In both the NAC-ABE scheme (utilizing the CP-ABE algorithm) and the proposed scheme (employing the enhanced CP-ABE algorithm), it was assumed that unmanned vehicles possessed $a + 2b$ attributes.

Table III outlines the cryptographic operations executed within each scheme. In comparison with the dummy scheme, both the NAC-ABE scheme and the proposed scheme showcased a diminished count of cryptographic operations within unmanned vehicles, attributed mainly to the subtask data reuse. Moreover, the proposed scheme demonstrated fewer cryptographic operations within the trusted control center (or

attribute authority) by virtue of streamlining the encryption process through the reuse of the same sub-policy and b attributes.

E. Scalability and Robustness

In the proposed fine-grained access control scheme, due to the adoption of the appropriate reuse of sub-policy, security cost is no longer directly proportional to the number of attributes in the whole access control policy. This to some extent enhances the scalability of the system in terms of attribute policy construction and CK encryption processes. Furthermore, the data reuse characteristics of ICN can also further mitigate the security cost at unmanned vehicles during task cooperation through efficient data reuse. It is noteworthy that, due to the absence of the need for real-time connected paths between communicating parties in ICN, the proposed scheme is more applicable to environments with intermittent connections among nodes in dynamic networks.

VIII. CONCLUSION

In this paper, we put forth an ICN-based secure task collaboration scheme aimed at counteracting eavesdropping attacks and unauthorized access attacks in challenging wireless edge networks. Our scheme initiates by introducing a secure task cooperation model, systematically quantifying the costs intertwined with task execution. Furthermore, we devise a tailored naming convention strategically poised to expedite collaborative task solicitations and the acquisition of requisite keys.

To ensure data confidentiality and access control for collaborative tasks with the same sub-policy, we incorporate attribute-based encryption into our scheme. This integration allows us to develop an enhanced name-based access control scheme, simplifying the encryption process for data reuse. Security analysis and experimental results show that our scheme significantly enhances task cooperation security at a low cost compared to existing schemes.

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Ningchun Liu received the B.E. degree from Dalian Jiaotong University, Dalian, China, in 2016, and the Ph.D. degree in communication and information systems from Beijing Jiaotong University, Beijing, China, in 2023. He is currently an Assistant Researcher with the Research Institute of Intelligent Networks, Zhejiang Laboratory. His research interests include information-centric Networks, software-defined networks, programmable data plane, and network security.



Guobiao He received the M.S. and Ph.D. degrees in communication and information systems from Beijing Jiaotong University in 2014 and 2021, respectively. He is currently with the National Computer Network Emergency Response Technical Team/Coordination Center of China, Beijing, China. His specific areas of research interest mainly focus on blockchain, cyber security, and future Internet architecture.



Shuai Gao (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in communication and information systems from Beijing Jiaotong University in 2001, 2004, and 2010, respectively. From 2015 to 2016, he was a Visiting Scholar with the University of Arizona. He is currently a Full Professor with the School of Electronic and Information Engineering, Beijing Jiaotong University. His research interests are in the areas of Internet architecture, sensor networks, and mobile Internet.



Hongke Zhang (Fellow, IEEE) received the M.S. and Ph.D. degrees in electrical and communication systems from the University of Electronic Science and Technology of China, Chengdu, China, in 1988 and 1992, respectively. He is currently a Professor with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing, China, and the Director of the National Engineering Research Center for Advanced Network Technologies, Beijing. He has authored more than ten books and the holder of more than 70 patents. His research has resulted in many papers, books, patents, systems, and equipment in the areas of communications and computer networks.



Xindi Hou is currently pursuing the Ph.D. degree with the National Engineering Research Center for Advanced Network Technologies, Beijing Jiaotong University. His currently research interests include clean-slate network architecture, future network addressing, and network integration.



Sajal K. Das (Fellow, IEEE) received the Ph.D. degree in computer science from the University of Central Florida, Orlando, FL, USA, in 1988. He is currently a Professor of Computer Science and a Daniel St. Clair Endowed Chair with the Missouri University of Science and Technology, Rolla, MO, USA. His extensive publications appeared in high quality journals and refereed conference proceedings. He holds five U.S. patents and coauthored four books. His H-index is 99 with more than 41 000 citations. His research interests include UAVs, wireless sensor networks, mobile and pervasive computing, cyber-physical systems and IoT, smart environments, cyber security, machine learning, and data science. He is the recipient of 12 best paper awards, the IEEE Computer Society Technical Achievement Award for pioneering contributions to sensor networks, and the University of Missouri System President's Award for Sustained Career Excellence. He is the Founding Editor-in-Chief of *Pervasive and Mobile Computing* (Elsevier), and an Associate Editor of the IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING, IEEE TRANSACTIONS ON MOBILE COMPUTING, and *ACM Transactions on Sensor Networks*.



Teng Liang received the B.S. degree from the Beijing Institute of Technology and the Ph.D. degree from the University of Arizona in 2020. He works with Peng Cheng Laboratory as an Associate Researcher. His research interests focus on named data networking, including forwarding, deployment, applications, and security in challenging scenarios. He also contributes code to the ICN/NDN community.